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EXERCISE SHEET / THEME 1: CATEGORICAL VARIABLES.

EXERCISE 1. TYPES OF VARIABLES.

Organize in a conceptual map the following concepts (some of them include others as subcategories). Also, think of concrete examples of these variables (i.e. a continuous variable could be the income of a worker expressed in Euros or its own country currency).

a) Binary or dummy variables.
b) Categorical variable (qualitative)
c) Continuous variables.
d) Discrete variables.
e) Nominal variables.
f) Ordinal variables

EXERCISE 2. EXTRACTING BASIC INFORMATION FROM A DATA BASE.

Context: You received a table with social information of 10 households. The table identifies the household with an ID, the gender of the household head, his/her employment situation, the household poverty situation, and finally the amount of people in the household that are within the working age (15-64 years old), and the amount of people that are not within the working age (i.e. younger and older).

Household id	Gender head	Employment head	Poverty situation	N working age	N non working age
1	Female	Unemployed	Extreme poverty	1	3
2	Female	Informal	Poverty	2	2
3	Female	Informal	No poverty	2	4
4	Female	Informal	Extreme poverty	1	3
5	Female	Formal	Poverty	2	2
6	Female	Formal	No poverty	1	3
7	Male	Unemployed	Extreme poverty	2	3
8	Male	Formal	Poverty	3	4
9	Male	Formal	No poverty	3	5
10	Male	Formal	No poverty	2	4

Using the given data set, answer the questions.

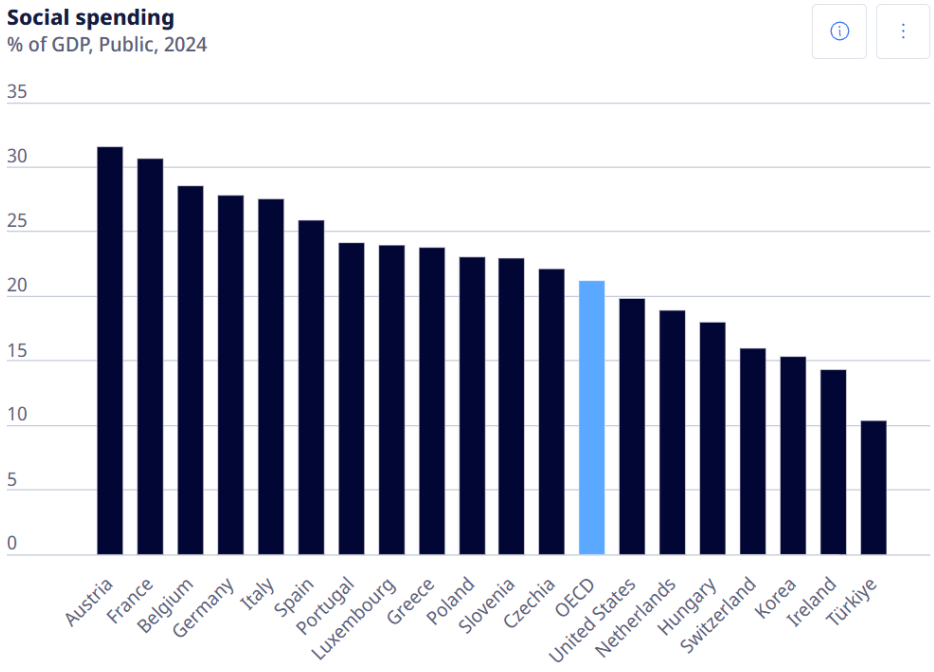
a) Identify types of variables (considering the kind of variables used in the Exercise 1).
b) Present the following:
    i. Bar plot of the employment status. (Hint: you can start with a frequency table of the employment status as an extra step if that is useful).

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ii. Frequency table of poverty situation (organized from most vulnerable to less vulnerable). Include at least number of observations, percentage and cumulative percentage.
iii. Estimate the mean of the total number of persons by household.

c) Present in bullet points the three most relevant take aways from your analysis.

EXERCISE 3. EXTRACTING BASIC INFORMATION FROM A GRAPH.



Source: OECD Data. Available at: https://www.oecd.org/en/data/indicators/social-spending.html

Considering the plot, answer the following questions:

a) What does the bar plot shows? What conclusions general ideas could you extract from it? (a good starting point could be identifying the kind of plot used, the variable that it displays, the year of the data, source of the information). If you want to go further, identify one or more groups of countries to mention something that you would like to highlight.
b) Considering the countries displayed in the bar plot, how many of them spend more than 20% of their GDP on social purposes?
c) Considering the countries displayed in the bar plot, which is the one with the lowest social spending? How much is that?

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d) Click on the link of the source (under the graph) and try to find the latest information available for Chile, Denmark, Peru and Sweden. Which is the percentage for each country? From which year? Could they be grouped?